

Splashgain: Empowering education sector with next-level edtech and assessment solutions with cloud scalability

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About Splashgain



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Splashgain achieved 99.9% uptime for its global services, gained a 30% boost to operational efficiencies, and supported up to 50,000 concurrent

Splashgain is an award-winning education technology company that provides digital assessment solutions, remote proctoring services, and onscreen evaluation services. The startup's mission is to be a leading global provider of innovative IT solutions that empower students and institutions by simplifying education processes. It serves more than 15 countries with over 500 customers and 10 million users. Striving to be among the top 10 edtech companies worldwide, Splashgain is committed to being present in the education life-

worldwide with G Kubernetes Engir

cycle of a student from Playgroup to Post-Graduation (PG to PG). It is a recipient of the PCQuest and Globee Business Awards.

Industries: Education, Technology

Location: India

Compute Engine

Google Cloud results

- Enables business growth and creative mission through 20% boost to operational efficiencies
- Supports global expansion by enabling platform stability in markets with low internet penetration
- Frees team to focus on solution building by reducing engineers needed for

100%
edtec
acros
marke

searche°

About Searce

Searce is one of the largest Google Cloud partners globally with over 12 years of experience, 150+ Google Cloud certifications and 300+ Google Cloud experts in-house. It works with digital native companies and enterprises to re-imagine and futurify their business.

Google Cloud (<https://cloud.google.com/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

Cloud Scheduler

(<https://cloud.google.com/scheduler/>)

Compute Engine

(<https://cloud.google.com/compute/>)

cloud
management
from 3 to 1

- Enables
investment
in solution
innovation
by reducing
operational
costs by 20%

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Vision AI (<https://cloud.google.com/vision/>)

Splashgain (<https://www.splashgain.com/>) is an award-winning Indian edtech platform that offers online examination services, remote test monitoring, and AI-enabled scoring and evaluation to academic institutions and multinational corporations around the world.

The company's reach extends to more than 15 countries, from advanced digital economies to emerging nations with low internet penetration, and it needs to maintain the same level of uptime and reliability during up to 50,000 concurrent tests. A single point of failure can invalidate an examination result, since uniform testing conditions are crucial to the fairness and accuracy of online exams.

The startup differentiates itself through AI-driven services such as automated evaluation of descriptive exam answers, which brings greater efficiency compared to manual assessments, while removing bias from the scoring process. It also offers 360-degree remote proctoring through

features such as facial and object recognition, and electronic device detection.

Splashgain began life on an on-premise architecture that deployed virtual machines (VMs) to power its testing, proctoring, and assessment operations. As the solution grew global, it encountered issues with platform uptime as well as the burden of manually configuring VMs for a growing workload. It switched to a major public cloud which improved its scalability and uptime on a VM infrastructure. But it still found itself falling short of the 99.99% uptime needed for an expanding global clientele.

With consulting partner [Searce](https://www.searce.com/) (<https://www.searce.com/>), Splashgain decided upon a full-fledged [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>) migration based on a [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine) (<https://cloud.google.com/kubernetes-engine>) (GKE) architecture to gain the system reliability and auto-scaling capabilities required to achieve its edtech mission. Splashgain was attracted by the planetary data center reach of Google Cloud, the same underlying infrastructure that powers Google Search. It also was drawn to GKE's built-in load balancing capabilities, which would enable stable testing services even in developing markets with slow internet connectivity.

"Our Google Cloud journey is exciting because it enables us to reach the farthest corners of the world, with confidence to deliver our services with zero downtime," says Swapnil Dharmadhikari, Founder and Director, Splashgain. "With Google Kubernetes Engine we're fast moving our entire

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Driving edtech empowerment at scale with Google Kubernetes Engine

Since migration to Google Cloud, Splashgain has automated 70% of its platform operations through GKE microservices, with 30% still dependent on manual VM provisioning on [Compute Engine](https://cloud.google.com/compute-engine) (<https://cloud.google.com/compute-engine>).

Dharmadhikari says the company is transitioning rapidly to move nearly all of its computing functions to a containerized GKE environment. He says this will bring it both greater platform agility and freedom from mundane chores for developers to focus on creative application building.

"Our immediate goal is to move to at least 95% platform automation on Google Kubernetes Engine," says Dharmadhikari. "That will mean less human intervention, less mistakes, more uptime guaranteed, and better delivery to global customers. In turn, we'll gain more opportunities for growth in exciting new markets."

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For app development, Cloud Functions (<https://cloud.google.com/functions>) is the serverless compute platform Splashgain employs for developers to rapidly release a continuous stream of AI-enabled features and upgrades. Meanwhile, Cloud Scheduler (<https://cloud.google.com/scheduler>) empowers the DevOps team, enabling clockwork automation for launching everything from application updates to big data analytics jobs.

In particular, Cloud Scheduler becomes instrumental for setting up concurrent exams, removing manual provisioning from the task of coordinating tens of thousands of tests around the world. "Using Cloud Scheduler, we can program cluster requirements required for each round of exams at a precise time with incredible ease," says Dharmadhikari.

Splashgain has achieved impressive outcomes in its edtech mission with a GKE-based architecture. Since migrating to Google Cloud, the platform has experienced zero downtime for more than 500 customers across all 17 countries where it operates. That compares to 4% downtime on its original on-premise architecture, and up to 2% downtime on the VM-based infrastructure of its legacy public cloud provider.

The ability to achieve 100% uptime on Google Cloud drives positive knock-on effects organization-wide, says Dharmadhikari. With any outage, company

divisions such as customer service, sales, and corporate communications come under pressure to address complaints and inquiries, as the engineering team races to fix the problem. With Google Cloud, these responsibilities have been all but eliminated, says Dharmadhikari.

Overall, the zero downtime that Google Cloud enabled helped the company achieve an estimated 20% boost to operational efficiency across its organization, thanks to factors such as reduction in customer service requests, and engineering tickets to fix problems.

That has resulted in a proportional 20% reduction in operational costs. Crucially, the number of technical experts required to manage Splashgain's cloud environment has gone down from three to one.

"Every department in our company is benefitting from the system stability, availability, and resilience enabled by Google Cloud," says Dharmadhikari.

"From customer service to human resources and engineering, Google Kubernetes Engine optimization is enabling us to drive a holistic picture of organizational transformation."

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**Leveraging proactive Searce
insight to optimize Google Cloud
load balancing performance**

According to Hrishikesh Galphade, Research and Development Manager at Splashgain, Searce played an instrumental role in Splashgain's Google Cloud journey through intensive sessions on best practices to optimize the GKE-based environment.

In particular, he says, Searce provided insight on optimizing GKE load balancing to enable the highest availability for microservices clusters across servers in the Google Cloud network. This was critical to ensuring reliability and scalability in regions where Splashgain gives tests that have poor internet connectivity.

"Even in markets that have internet speeds of under one mbps, we are able to offer exams with zero downtime, under our maximum target load time of three seconds, thanks to the auto-scaling power and built-in load balancing features of Google Kubernetes Engine," says Galphade. "Searce provided key insights on how to optimize GKE load balancing capabilities."



Swapnil Dharmadhikari, Founder and Director, Splashgain

Protecting sensitive education data through state-of-the-art security features

For an edtech solution, Splashgain needs to be vigilant about security and data privacy to comply with global regulations and protect the personal information it handles from global clients. Google Cloud's built-in data protection protocols enable Splashgain to achieve rock-solid security for the

data of test-takers, educational institutions and corporations around the world.

In particular, Google Cloud's multi-layer encryption across a distributed data storage protocol gives Splashgain the peace of mind it needs to operate securely in the global marketplace. Meanwhile,

Cloud Backup and DR Service

(<https://cloud.google.com/solutions/backup-dr>) provides Splashgain the seamless data protection and business continuity it needs in the event of a security breach or DoS attack.

"We rely heavily on Google Cloud advanced security mechanisms, including disaster recovery," says Dharmadhikari. "If an attack happens, we have the peace of mind that we have a world class mechanism in Cloud Backup and DR Service to restore data and services in real time."

Accelerating the future of global edtech with AI-driven insight

Looking into the future, Splashgain is excited to continue developing advanced AI solutions to improve accuracy of testing, platform automation, and bias reduction in the examination process. To help drive this AI mission, Splashgain is moving to adopt BigQuery (<https://cloud.google.com/bigquery>) as its data warehouse to power machine-learning enabled models for more sophisticated AI-driven exam generation and evaluation.

It is also actively deploying Vertex AI (<https://cloud.google.com/vertex-ai>) and Vision AI

(<https://cloud.google.com/vision>) to develop remote test monitoring protocols for greater fairness and accuracy in proctoring services. For example, it is training proprietary optical character recognition models with these solutions to confirm the authenticity of handwritten answers in a test document.

"From the seamless auto scaling capabilities enabled by Google Kubernetes Engine to Google AI tools such as Vision AI, we see unlimited potential to develop new creative edtech solutions and spread our mission of fair, effective, and empowering testing in new markets around the world," says Dharmadhikari.